

KAIRO: The Engineering Intelligence Platform

Subtitle: Existing platforms optimize workflows. KAIRO optimizes engineering decisions.

1. The Problem: The Architecture Gap

Modern software engineering relies on highly distributed toolchains (GitHub, Slack, Jira, Notion). While these tools efficiently track the current status of tasks, they inherently fail to preserve the underlying reasoning behind critical engineering decisions. Over time, the intended architecture diverges from the actual codebase. Recurring bottlenecks remain hidden in fragmented chat logs, leading teams to repeatedly solve the same problems without understanding their root causes. The industry lacks a cohesive engineering memory.

2. Proposed Solution: The Engineering Memory Engine™

KAIRO is an Engineering Intelligence Platform powered by a **Temporal Knowledge Graph**. Every engineering artifact—commit, pull request, Slack discussion, issue, documentation update, and design decision—is converted into entities and relationships inside a temporal knowledge graph. AI agents reason over this evolving graph to reconstruct engineering rationale and identify recurring patterns.

3. Novel Research Contribution & Metrics

While existing research in AI project management focuses on document reasoning (standard Vector RAG), KAIRO pioneers **Software Engineering Reasoning**. To quantify longitudinal project health, KAIRO introduces two novel algorithms:

- **Decision Drift Index (DDI):** A mathematical measure of how far the current codebase implementation has drifted from the original architectural intent.
- **Engineering Evolution Score (EES):** A continuous metric that evaluates how architectural decisions, developer interactions, and code changes influence overall project stability and technical debt over time.

4. Existing Solutions vs. KAIRO

Feature / Platform	GitHub Projects	Jira	KAIRO (Our Solution)
Core Paradigm	Basic issue tracking	Complex workflow management	Project Intelligence & Reasoning
Context Maintenance	Outdated project context	Heavy manual updates required	Automated via Webhooks & Graph
Historical Reasoning	None	None	Engineering Memory Engine™

5. The Architectural Flow

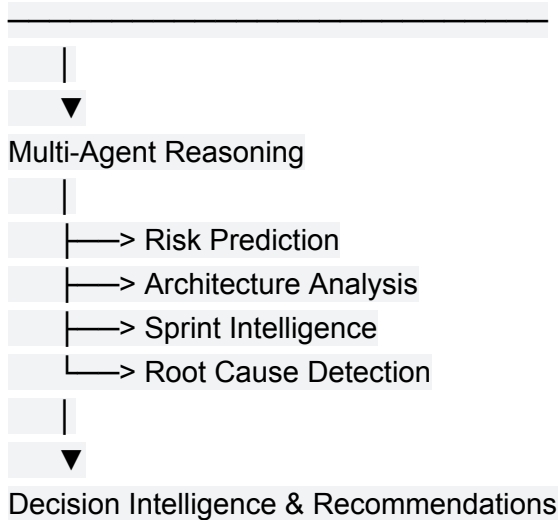
Engineering Events (Commits, PRs, Slack, Meetings, Issues, CI/CD)



Knowledge Extraction Layer



Engineering Memory Engine™
(Temporal Graph Builder)



6. System Architecture & Tech Stack

- **Frontend:** Next.js (React), TypeScript, Tailwind CSS, shadcn/ui.
- **Backend API:** FastAPI (Python) for stateless, high-performance integration with AI workflows.
- **Databases:** PostgreSQL (Relational), pgvector (Vector), Neo4j (Graph - Core).
- **Intelligence Layer:** Multi-Agent framework (LangGraph) orchestrating LLMs over a Temporal GraphRAG architecture.
- **Event Pipeline:** Webhooks streaming into a lightweight message queue (Redis Pub/Sub) to dynamically update the Neo4j graph without blocking the main API.

7. Academic Research Integration

Research Topic: *Engineering Memory: A Temporal Knowledge Graph Framework for Longitudinal Software Project Intelligence*

Brief Info: Current AI project management relies heavily on standard Retrieval-Augmented Generation (RAG) using vector databases, which suffer from "temporal blindness" and fail to execute multi-hop relational reasoning. This research explores how constructing a time-stamped Knowledge Graph (using Neo4j) to ingest disparate artifacts (code commits, pull requests, developer discussions) allows AI to accurately trace the origin of architectural shifts. By proving that Temporal GraphRAG significantly outperforms baseline vector RAG in root-cause analysis, this project contributes a novel framework for automating the reconstruction of lost engineering

rationale.

Core References & Literature Context

- **From Local to Global: A Graph RAG Approach to Query-Focused Summarization** (Darren Edge, Ha Trinh, et al., Microsoft Research, 2024)
Application: Foundational citation establishing how shifting from flat text chunks to structured knowledge graphs enables multi-hop AI reasoning.
- **Knowledge Graph Enhanced Retrieval-Augmented Generation for Failure Mode and Effects Analysis** (Bahr et al., Fraunhofer, 2024)
Application: Directly aligns with KAIRO's risk agents, proving that combining Graph Queries with Vector Search successfully retrieves diagnostic information for root-cause analysis.
- **Enhancing Performance of LLM-Based Problem-Solving RCA Chatbot in Technical Investigations** (2024)
Application: Supports the core argument that graph-guided retrieval drastically improves contextual grounding over baseline vector retrieval in noisy technical data.
- **A Comprehensive Survey on Root Cause Analysis in (Micro) Services: Methodologies, Challenges, and Trends** (Tingting Wang, Guilin Qi, 2024)
Application: Highlights the severe limitations of relying on isolated symptomatic data in interconnected software architectures without understanding propagative dependencies.